CLASS 9 · CBSE · GANITA MANJARI PART 2 · SHORT NOTES

Chapter 9: Propositions and their Converses

Topic-wise short notes with solved examples

A proposition is a statement that is either true or false. This chapter looks at "if … then …" statements, their converses, and how one counterexample is enough to show that a statement is false.

1 Propositions

- A **proposition** is a statement that is either true or false.
- Many propositions have the form **"If X, then Y"**. This is also written "X implies Y" or "Y when X". All three mean the same.
- X is the condition (hypothesis) and Y is the conclusion.

Example 1.1 Write "A number has an odd number of factors when it is a perfect square" in the form "If X, then Y".

SOLUTION

The condition is "the number is a perfect square". The conclusion is "it has an odd number of factors".

If a number is a perfect square, then it has an odd number of factors.

2 The converse

- The **converse** of "If X, then Y" is "If Y, then X". The condition and the conclusion change places.
- Any opening sentence that sets the scene (such as "Let a, b, c be the sides of a triangle") stays as it is.
- The truth of a proposition tells us **nothing** about its converse. Each must be checked separately.

Proposition	Converse	Example
True	True	If a triangle is equilateral, then each of its angles is 60°.
True	False	If a number is a multiple of 6, then it is a multiple of 3.
False	True	If $ab = 0$, then $a = 0$.
False	False	If a number is prime, then it is odd.

Example 2.1 Write the converse of "If it rains, then the road is wet". Is the converse true?

SOLUTION

Converse: "If the road is wet, then it has rained."

The proposition is true, but the converse need not be. A water tanker may have spilled water on the road.

3 Counterexamples

- A **counterexample** is one example in which the condition holds but the conclusion fails.
- **One** counterexample is enough to show that a proposition is false. No number of supporting examples can prove it true: for that, a general argument is needed.

Example 3.1 Show that "If a number is a multiple of 3, then it is a multiple of 6" is false.

SOLUTION

9 is a multiple of 3 but not a multiple of 6.

9 is a counterexample, so the statement is false.

Example 3.2 Find a counterexample to "All numbers of the form $n^2 + n + 11$ are prime".

SOLUTION

For $n = 1, 2, 3, \ldots, 9$ the values 13, 17, 23, 31, 41, 53, 67, 83, 101 are all prime.

For $n = 10$: $100 + 10 + 11 = 121 = 11 \times 11$.

$n = 10$ is a counterexample.

REMEMBER

Fermat believed that every number of the form $2^{2^n} + 1$ is prime. The first five (3, 5, 17, 257, 65537) are prime, but Euler showed that the next one, $2^{32} + 1 = 4294967297$, equals 641×6700417. No simple formula is known that gives only primes.

4 When a proposition and its converse are both true

- If "If X, then Y" and "If Y, then X" are both true, then X and Y **imply each other**. Each holds exactly when the other does.
- **Perfect squares and odd numbers of factors.** The factors of a number come in pairs whose product is the number: for 12 the pairs are $(1, 12)$, $(2, 6)$, $(3, 4)$.
- If n has an odd number of factors, some factor must be paired with itself, so $n = f \times f$ is a perfect square.
- If n is a perfect square, exactly one factor is paired with itself and the rest come in pairs, so the number of factors is odd.
- So both statements are true: n is a perfect square **exactly when** it has an odd number of factors.

Example 4.1 List the factors of 36 and of 20 and check the statement above.

SOLUTION

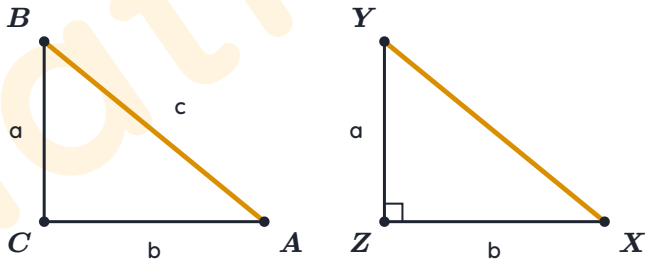
36: 1, 2, 3, 4, 6, 9, 12, 18, 36. That is 9 factors, an odd number, and $36 = 6^2$.

20: 1, 2, 4, 5, 10, 20. That is 6 factors, an even number, and 20 is not a perfect square.

Both agree with the statement.

5 Proving both directions

- The two directions need two arguments. An argument that starts from Y and reaches X proves only the converse.
- **Isosceles triangles.** Proposition: if two sides of a triangle are equal, then the angles opposite them are equal. Converse: if two angles of a triangle are equal, then the sides opposite them are equal. Both are true.
- **Baudhāyana–Pythagoras theorem.** Let a, b, c be the sides of a triangle. If the triangle is right-angled (with hypotenuse c), then $a^2 + b^2 = c^2$.
- **Its converse.** Let a, b, c be the sides of a triangle. If $a^2 + b^2 = c^2$, then the triangle is right-angled.
- Proof of the converse: draw a right-angled triangle XYZ whose perpendicular sides are a and b. By the theorem its hypotenuse is $\sqrt{a^2 + b^2} = c$. So the two triangles have the same three sides and are congruent by SSS. Hence the given triangle has a right angle too.



Triangle ABC has a² + b² = c². Triangle XYZ is drawn with a right angle at Z.

Example 5.1 A triangle has sides 9 cm, 40 cm and 41 cm. Is it right-angled?

SOLUTION

$9^2 + 40^2 = 81 + 1600 = 1681 = 41^2$

Yes, by the converse of the Baudhāyana–Pythagoras theorem. The right angle is opposite the 41 cm side.

6 More examples to study

Proposition	True?	Converse true?
If two lines are parallel, then corresponding angles made by a transversal are equal.	Yes	Yes
If a quadrilateral is a square, then all its angles are equal.	Yes	No (a rectangle)
If two triangles are congruent, then they have the same area.	Yes	No
If $x = y$, then $x^2 = y^2$.	Yes	No ($x = 3, y = -3$)
If $x = y$, then $x^3 = y^3$.	Yes	Yes
If n is divisible by 24, then it is divisible by 4 and by 6.	Yes	No ($n = 12$)
If n is divisible by 60, then it is divisible by 5 and by 12.	Yes	Yes

- The last two rows differ because 5 and 12 have no common factor, while 4 and 6 share the factor 2.

Example 6.1 To test whether a number is divisible by 8, is it enough to check that it is divisible by 2 and by 4?

SOLUTION

"If a number is divisible by 8, then it is divisible by 2 and by 4" is true.

The test needs the converse, which is false: 12 is divisible by 2 and by 4 but not by 8.

No. A divisibility test works only when the converse is true.

7 Common mistakes

- Assuming the converse is true because the proposition is true.
- Trying to prove a statement with a few examples. Examples can only disprove.
- Giving a "counterexample" in which the condition does not hold. In a counterexample the condition is true and the conclusion is false.
- Changing the opening sentence when writing the converse. Only the "if" part and the "then" part are exchanged.
- Proving "If Y, then X" and claiming to have proved "If X, then Y".